Week 6 — For Loops & Patterns

Name: ____ Date: ____

> *Colon chant: **first : difference : last — and the last one COUNTS.***

Today's words

Word	What it means
`for` **loop**	repeats a block a set number of times, closed by `end`
loop variable	the lap counter (`i`) — changes every lap
colon sequence `start:step:stop`	deals numbers: first term, jump size, last term — last one **included**
accumulator	a snowball variable: `total = total + n`
`repmat('*', 1, n)`	the repeater — n copies of the star, glued together

1 · Match the sequence to its colon 🔗

Draw a line from each sequence to the colon expression that makes it. (One expression is a fake — it makes NOTHING: an empty sequence!)

Sequence		colon expression
1, 2, 3, 4, 5		`2:2:10`
2, 4, 6, 8, 10		`1:5`
5, 10, 15, 20		`10:-1:1`
10, 9, 8, … 1		`5:1:1`
		`5:5:20`

2 · Be the computer: trace the snowball 🤖

```
total = 0;
for n = 1:4
    total = total + n;
end
disp(total)
```

Fill in the table, one lap per row, then the final line:

lap	`n`	`total` after the lap
1		
2		
3		
4		

The screen shows: ______

3 · What prints? ⭐

```
for i = 1:3
    disp(repmat('*', 1, i))
end
```

Draw the exact output:

🧠 Brain teaser (optional — take it home)

Gauss beat his teacher by pairing numbers: 1+100, 2+99, … = 50 pairs of 101 = **5050**.

Use his trick — no computer, no calculator — to add 1 + 2 + 3 + … + 1000. How many pairs? What does each pair add up to? Then the showdown question: which colon sequence would make a loop check your answer — `1:1000` or `0:999` — and *why does the other one get it wrong?* (Both deal exactly 1000 numbers…) Bring your answer next week for a shout-out.